

# Customer **Churn** Intelligence & Retention System

*From prediction to action: transforming customer data into retention strategies*

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Machine Learning Project



# Business Context

*Why problem exists*

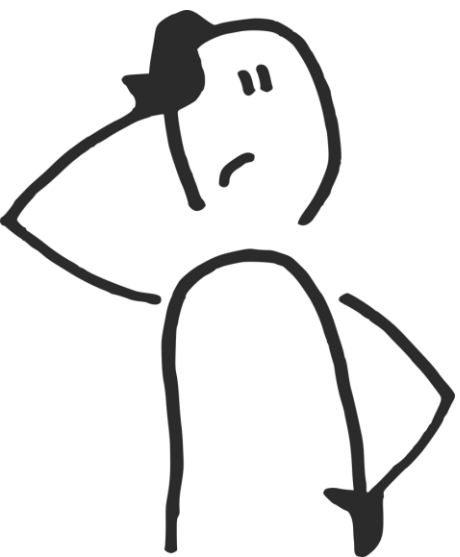


- High customer **churn** impacts revenue growth.
- **Retention** is cheaper than acquisition.
- Identifying **at-risk customers** is challenging.
- Customer data contains hidden **behavioral patterns**.
- Businesses need intelligent retention **solutions**.

# Problem Statement

*What exactly is the problem*

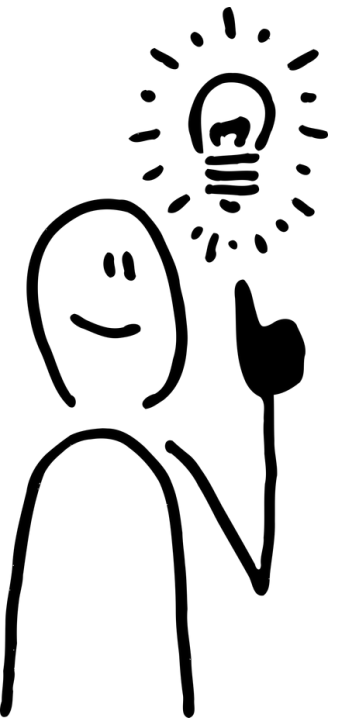
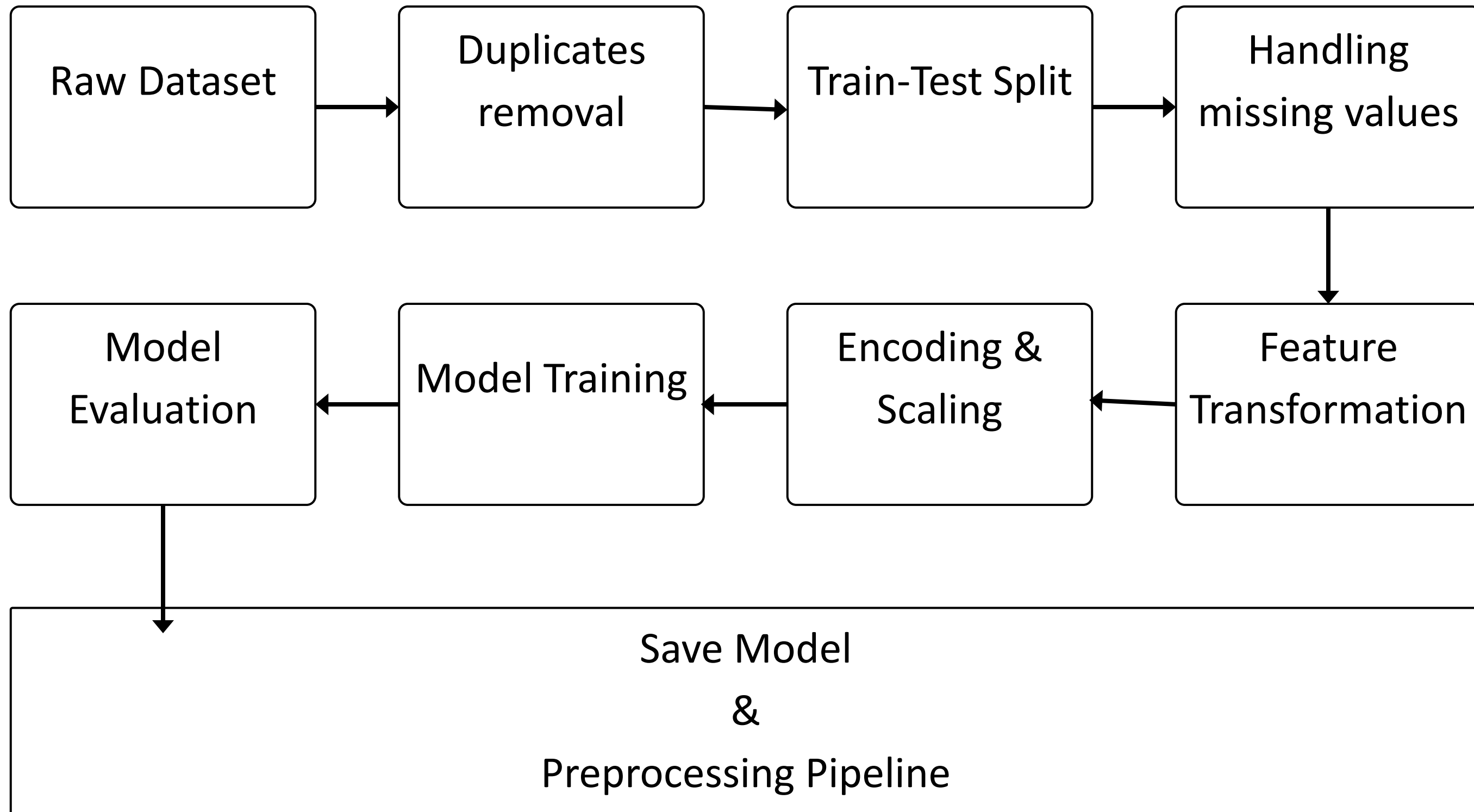
- Customers don't leave suddenly — their behavior changes slowly.
- These early signs are often missed.
- Businesses realize only after the customer has already left.
- There is no clear way to identify and focus on risky customers.
- So, there is a need for a smart system to predict and prevent churn.



# Approach

## *Model Development*

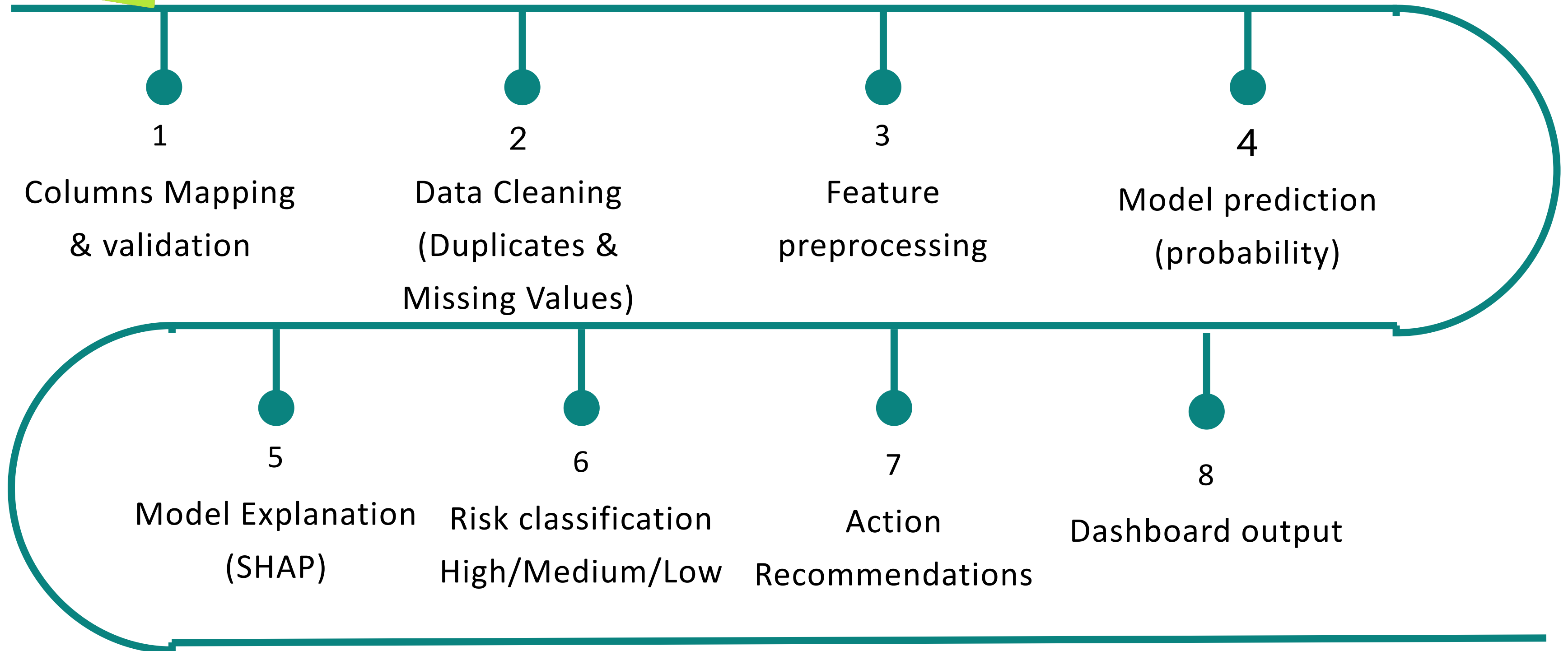
### *Pipeline*





# System Workflow

*How the model works with new customer data*



# UI/Dashboard (Demo)



- ✓ Column validates.....
- ✓ Duplicated removed: 120
- ✓ Missing values handled: 85
- ✓ Preprocessing applied successfully

Total customers  
\$12M



Predicted High  
Risk %



from dataset

High risk customers



Medium risk  
customers



Low Risk customers



Revenue at Risk  
\$345M



67%  
67%  
67%

Risk

▼

- High
- Medium
- Low

Action

▼

- Discount
- Email
- Reward

| ID | Churn Probability | Revenue at risk | Top drivers                       | Segmentation | Recommended action |
|----|-------------------|-----------------|-----------------------------------|--------------|--------------------|
| 10 | 0.82              | 48000           | High inactivity, cart abandonment | High         | Discount           |
| 25 | 0.56              | 22000           | Low engagement                    | Medium       | Email campaign     |
| 39 | 0.18              | 10000           | High activity                     | Low          | Loyalty reward     |



Download  
Filtered CSV



Download  
CSV